

Exogenous Reservoir Volatility with an Exponential-Quadratic Readout

Current specification, physical stylised facts, and joint SPX–VIX calibration architecture

Purpose. This document consolidates the current candidate descended from ESN–A2. It states the model under the physical and pricing measures, identifies the mechanisms supporting each targeted stylised fact, and separates mathematical implications from numerical evidence and open calibration questions.

Status: research specification, not a calibration result

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Abstract

The objective is a continuous-path stochastic-volatility model that can reproduce, under the physical measure, finite-frequency heavy tails, volatility clustering, roughness, leverage, a strong Zumbach effect and, at lower priority, the Taylor effect. Under the pricing measure, the same structural model should be capable of fitting SPX and VIX option surfaces. The computational constraint is that changes in readout and risk-premium parameters must not require new simulation of the exogenous state paths.

The proposed model uses a structured bank of Ornstein–Uhlenbeck factors as a linear reservoir, an optional bounded nonlinear echo-state layer, and a normalised exponential-quadratic variance readout. A shifted quadratic projection of a return-correlated but variance-independent OU bank supplies leverage and time-reversal asymmetry. Slow exogenous OU modes supply clustering; a rough-kernel lift supplies finite-scale roughness; a fast, partly return-correlated direction supplies VIX-like spikes; and an independent quadratic block is reserved for dispersion or tail tuning. A deterministic Girsanov kernel preserves Gaussianity of the OU core between $\mathbb{P}$ and $\mathbb{Q}$.

The architecture supports all requested mechanisms without a quintic polynomial. This is a capacity statement, not a proof that a single parameter vector fits every physical statistic and both option surfaces simultaneously. The main unresolved questions are empirical joint calibration, identifiability, and the placement of the upper tempering level. With bounded variance, true stock martingality follows on finite horizons by Novikov; without tempering, the exponential-quadratic moment and martingale domains must be enforced explicitly.

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1 Objective and standard of claim

Let S denote the total-return-adjusted equity index level and V its instantaneous variance. The programme has two linked objectives:

1. Under $\mathbb{P}$, reproduce the relevant empirical properties of equity returns and volatility at intraday and daily frequencies.
2. Under $\mathbb{Q}$, reproduce the cross section of SPX options, VIX futures and VIX options, while reusing the same exogenous random paths during calibration.

The phrases *supports* and *reproduces* must not be conflated. Four evidence levels are used throughout:

Label	Meaning	Interpretation
C	By construction	Follows directly from the state or readout definition.
P	Conditional proposition	Follows under explicit non-degeneracy, moment or scale assumptions.
N	Numerical mechanism evidence	Demonstrated by an ablation or controlled simulation, but not yet by market calibration.
O	Open empirical question	Must be established on historical data or option surfaces.

Main conclusion

There is no known structural incompatibility between the requested physical-measure facts, path reuse, and joint SPX–VIX pricing in the proposed class. There is also no theorem that a successful joint fit must exist. The correct conclusion is architectural feasibility subject to explicit moment, martingale and calibration tests.

2 The current model

2.1 Driver system and measure convention

Work on a filtered probability space satisfying the usual conditions. Under a measure $\mathbb{M} \in \{\mathbb{P}, \mathbb{Q}\}$, let $W^{S, \mathbb{M}}$ drive the equity return. The structured reservoir uses four driver groups:

$$dW_t^{F, \mathbb{M}} = \rho_F dW_t^{S, \mathbb{M}} + \sqrt{1 - \rho_F^2} dB_t^{F, \mathbb{M}}, \quad (1)$$

$$dW_t^{J, \mathbb{M}} = \rho_J dW_t^{S, \mathbb{M}} + \sqrt{1 - \rho_J^2} dB_t^{J, \mathbb{M}}, \quad (2)$$

where the superscripts F and J stand for feedback and fast-spike banks. The clustering drivers B^C and optional orthogonal drivers B^O are independent of W^S, B^F, B^J . Partial rather than zero correlation is recommended for the fast bank, because large VIX moves are empirically associated with equity sell-offs.

2.2 Layer 1: structured OU reservoir

For each bank $b \in \{F, J, C, O\}$, define under $\mathbb{P}$ the OU coordinates

$$dX_{j,t}^{b,\mathbb{P}} = -\lambda_j^b X_{j,t}^{b,\mathbb{P}} dt + \sqrt{2\lambda_j^b} dW_{j,t}^{b,\mathbb{P}}, \quad j = 1, \dots, N_b. \quad (3)$$

Within the feedback bank, the coordinates share W^F ; this creates the correlated multi-rate lift used to approximate a rough Volterra kernel. The fast bank can share one or a small number of partially return-correlated drivers. The clustering and orthogonal banks may use independent drivers, so that their covariance is a flexible positive mixture of exponentials.

For $N_b \geq 2$, the rate grids are geometric and fixed in the inner calibration:

$$\lambda_j^b = \lambda_{\min}^b \left(\frac{\lambda_{\max}^b}{\lambda_{\min}^b} \right)^{\frac{j-1}{N_b-1}}. \quad (4)$$

Define normalised projections

$$Y_t = a^\top X_t^F, \quad F_t = f^\top X_t^J, \quad \text{Var}_{\mathbb{P}}(Y_t) = \text{Var}_{\mathbb{P}}(F_t) = 1 \quad (5)$$

at the stationary physical reference point. The direction Y carries the main leverage–Zumbach mechanism; F controls the front end of volatility spikes.

2.3 Layer 2: optional bounded echo-state layer

On a time grid $t_n = n\Delta t$, with $0 < \alpha \leq 1$, the nonlinear reservoir is

$$Z_{n+1} = (1 - \alpha)Z_n + \alpha \tanh(AZ_n + B\psi(X_n) + b), \quad (6)$$

where A, B, b are drawn or designed once and then held fixed. The feature map ψ may contain selected linear, quadratic and cross-bank features. A sufficient contraction condition for the echo-state property is

$$(1 - \alpha) + \alpha \|A\|_2 < 1. \quad (7)$$

Because Z is driven only by the exogenous OU state, neither X nor Z changes when a readout coefficient changes. The recommended first calibration sets the nonlinear loading δ below to zero. It is activated only if a physical-measure ablation demonstrates an incremental gain that justifies losing Gaussian conditional-moment formulas.

Terminology

With $\delta = 0$, the operative model is a structured Gaussian reservoir rather than a nonlinear ESN in the narrow, classical sense. It remains an ESN-style fixed-reservoir model. The full two-layer ESN is obtained when $\delta^\top Z_t$ enters the readout.

2.4 Exponential-quadratic readout

The multiplier-free log-variance score is

$$\eta_t = c_C^\top X_t^C + q_Z(Y_t - \beta)^2 + q_J(F_t - \beta_J)^2 + \frac{q_O}{N_O} \|X_t^O\|^2 + \delta^\top Z_t. \quad (8)$$

The recommended core imposes $q_Z > 0$, $q_J \geq 0$, $\beta_J \geq 0$, and starts at $q_O = 0$, $\delta = 0$. Setting $\beta_J = 0$ exactly recovers the original Stage-3 M2 readout. Each shifted quadratic is deliberately parameterised without an additional linear loading on the same projection: adding both ℓF and $q_J(F - \beta_J)^2$, for example, would make the linear loading and shift redundant up to deterministic normalisation.

For a cap level L and sharpness $\kappa > 0$, define the smooth upper tempering function

$$h_{L,\kappa}(x) = L - \frac{1}{\kappa} \log\left(1 + e^{\kappa(L-x)}\right), \quad G_{L,\kappa}(x) = e^{2h_{L,\kappa}(x)}. \quad (9)$$

It satisfies $h_{L,\kappa}(x) \approx x$ well below L and $h_{L,\kappa}(x) \uparrow L$ as $x \rightarrow \infty$. Let $\xi_0(t)$ be the market forward variance curve and let $\underline{v}$ be a small deterministic variance floor. The variance is

$$V_t = \underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t)}{M_{\mathbb{Q}}(t)}, \quad M_{\mathbb{Q}}(t) = \mathbb{E}_0^{\mathbb{Q}}[G_{L,\kappa}(\eta_t)]. \quad (10)$$

Provided $\xi_0(t) > \underline{v}$, positivity and $\mathbb{E}_0^{\mathbb{Q}}[V_t] = \xi_0(t)$ follow by construction. The untempered shadow model is obtained by setting $h(x) = x$; it is useful for analytic checks and control variates, but it is not the recommended production specification on a return-correlated quadratic bank.

2.5 Equity dynamics under $\mathbb{P}$ and $\mathbb{Q}$

Under the physical measure, use a deterministic instantaneous Sharpe ratio $\theta(t)$:

$$\frac{dS_t}{S_t} = (r_t - d_t + \theta(t)\sqrt{V_t}) dt + \sqrt{V_t} dW_t^{S,\mathbb{P}}. \quad (11)$$

Let the full Brownian vector satisfy the deterministic Girsanov shift

$$dB_t^{\mathbb{P}} = dB_t^{\mathbb{Q}} - \gamma(t) dt, \quad \gamma_S(t) = \theta(t). \quad (12)$$

Then

$$\frac{dS_t}{S_t} = (r_t - d_t) dt + \sqrt{V_t} dW_t^{S,\mathbb{Q}}. \quad (13)$$

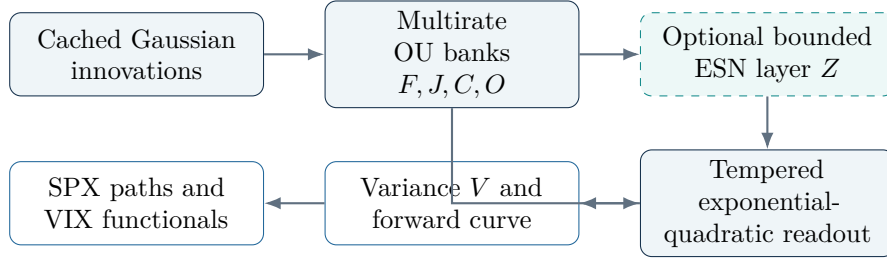
The unspanned components of γ are parsimonious variance-risk-premium parameters. Because γ is deterministic, every OU bank remains Gaussian under $\mathbb{Q}$: only its deterministic conditional mean changes, while its covariance and rate spectrum are preserved. This is the essential reason the same centred Gaussian innovations can be reused under both measures.

More explicitly, if $\gamma_b(t)$ denotes the effective deterministic shift of the Brownian driver of coordinate (b, j) , then

$$dX_{j,t}^{b,\mathbb{Q}} = \left(-\lambda_j^b X_{j,t}^{b,\mathbb{Q}} - \sqrt{2\lambda_j^b} \gamma_b(t)\right) dt + \sqrt{2\lambda_j^b} dW_{j,t}^{b,\mathbb{Q}}. \quad (14)$$

For the correlated banks, γ_b is the corresponding linear combination of the equity and unspanned primitive-driver premia.

2.6 Architecture at a glance



3 Parameter hierarchy and path reuse

3.1 Hyperparameters, structural parameters and daily inputs

Class	Examples	Treatment
Architectural hyperparameters	Bank dimensions; OU rate grids; reservoir matrices A, B ; leak α ; feature map ψ ; cap family	Selected in an outer validation loop and changed infrequently. Changing OU rates or fixed reservoir weights requires rebuilding state features.
Shared structural parameters	$a, f, c_C, q_Z, \beta, q_J, q_O, \delta, \rho_F, \rho_J$	Estimated from a panel of physical and option data with regularisation. Readout changes do not change cached states; correlation changes require rebuilding correlated states from the same primitive innovations.
Measure-dependent parameters	θ and unspanned components of γ	Thin deterministic parameterisation. They shift conditional means but do not change OU covariances.
Daily inputs and states	$\xi_0(\cdot)$, rates, dividends, spot, filtered X_t, Z_t	Updated daily. They are inputs or filtered states, not structural calibration parameters.

3.2 What path reuse does and does not mean

For fixed hyperparameters and correlations, simulate or reconstruct once:

$$\{\Delta W^S, \Delta B^F, \Delta B^J, \Delta B^C, \Delta B^O; X; Z\}. \quad (15)$$

When $q_Z, \beta, q_J, q_O, c_C, \delta$ or the deterministic risk premia change, no new random numbers are required. The readout, variance, stock integral and payoffs must nevertheless be recomputed. It is impossible for the final S or VIX paths to remain unchanged when the volatility readout changes.

Change	New random draw?	Required recomputation
Readout coefficient	No	η, V, S , VIX map and payoffs
Deterministic risk premium	No	Deterministic OU mean shift; possibly Z ; then pricing quantities
Forward-variance curve	No	Variance level, S , VIX and payoffs
Correlation	No, if primitive shocks are cached	Correlated drivers and affected OU states
OU rate grid or reservoir matrix	No new seed is necessary, but the state changes	Full feature-engine reconstruction

This is the same computational principle exploited by precomputed signature and quantisation approaches: expensive stochastic input is generated offline, while parameter-dependent maps are evaluated during optimisation [4, 2].

4 Physical-measure stylised facts

The selected targets are standard empirical regularities of equity markets, with roughness and time-reversal asymmetry adding more discriminating requirements to the classical inventory [3, 5, 8].

4.1 Negligible linear autocorrelation of returns

For non-overlapping increments over a short interval Δ , the martingale part of equation (11) has conditional mean zero and variance of order Δ . The predictable return component is of order Δ , so its contribution to return covariance is of order Δ^2 . Consequently, linear return autocorrelation is small at intraday and daily frequencies unless the risk-premium term is made too large or too state dependent.

Under $\mathbb{Q}$, discounted simple gains are martingale increments and therefore orthogonal across non-overlapping intervals. Log returns need not be exactly uncorrelated because their drift contains $-V_t/2$.

Status: P/O. Near-zero autocorrelation is structurally natural, not an identity under $\mathbb{P}$. It must remain an explicit diagnostic, especially after estimating θ .

4.2 Volatility clustering

The recommended architecture obtains clustering from the exogenous slow bank rather than the endogenous activity filter. If $C_t = c_C^\top X_t^C$ and the coordinates are independent stationary OU processes, then

$$\text{Cov}(C_t, C_{t+\tau}) = \sum_{j=1}^{N_C} c_{C,j}^2 e^{-\lambda_j^C \tau}. \quad (16)$$

A geometrically spaced rate grid approximates slowly decaying covariance over a prescribed finite interval. The monotone exponential link transfers persistence from η to V ; quadratic

channels add modes with decay rates such as $2\lambda_j$ and $\lambda_i + \lambda_j$.

This choice preserves strong frozen-state reuse. An endogenous activity bank

$$d\Sigma_{k,t} = \alpha_k(V_t - \Sigma_{k,t}) dt \quad (17)$$

remains an optional challenger if the exogenous bank cannot fit the empirical volatility ACF. It needs no new random numbers, but it must be recomputed whenever the readout changes.

Status: C/P/O. Persistence is built into the rate mixture; matching its empirical magnitude and shape is a calibration question.

4.3 Finite-scale roughness

Rough volatility is usually expressed through the small-lag scaling

$$\mathbb{E}[(\log V_{t+\tau} - \log V_t)^2] \asymp C\tau^{2H}, \quad H < \frac{1}{2}, \quad (18)$$

with empirical estimates often near $H = 0.1$ [5]. A fractional kernel can be approximated on a chosen range by a positive exponential sum,

$$K_H(t) \approx \sum_{j=1}^{N_F} a_j e^{-\lambda_j^F t}, \quad (19)$$

which is precisely a Markovian OU lift [1]. The feedback projection $Y = a^\top X^F$ therefore reproduces equation (18) over the frequencies covered by the rate grid.

If g is the smooth map from the state to log variance, then

$$g(Y_{t+\tau}) - g(Y_t) = g'(Y_t)(Y_{t+\tau} - Y_t) + o_{\mathbb{P}}(\tau^H). \quad (20)$$

Thus the exponent is locally preserved whenever $g'(Y_t) \neq 0$. The derivative can vanish at the parabola vertex and becomes small near cap saturation, so roughness should be estimated on the bulk of the volatility distribution.

Finite lift, finite claim

Every finite OU system is a semimartingale and has Brownian $H = 1/2$ behaviour at sufficiently small asymptotic lags. The defensible claim is rough behaviour on the intraday-to-daily scale range selected by $[\lambda_{\min}^F, \lambda_{\max}^F]$, not genuine fractional asymptotics at arbitrarily small times.

Status: P/O. The lift supports the target scaling over a finite band; the realised band and estimated H must be verified with a noise-robust realised-volatility estimator.

4.4 Leverage effect

For a lag $\tau > 0$, define the leverage statistic

$$L(\tau) = \text{Corr}_{\mathbb{P}}(r_t, V_{t+\tau}), \quad L(\tau) < 0 \text{ empirically.} \quad (21)$$

The feedback contribution is $q_Z(Y - \beta)^2$, with derivative

$$\frac{\partial \eta}{\partial Y} = 2q_Z(Y - \beta). \quad (22)$$

Near the stationary centre $Y \approx 0$, its sign is $-\text{sign}(q_Z\beta)$. Since the outer link is increasing away from cap saturation, the instantaneous return–log-variance covariation has leading sign

$$\text{sign}(\text{d}(\log S, \log V)_t) \approx -\text{sign}(q_Z\beta\rho_F). \quad (23)$$

Under the present convention, $q_Z > 0$, $\beta > 0$, $\rho_F > 0$ therefore produce negative leverage. This apparently unusual positive correlation is a consequence of the shifted parabola convention; reversing both β and ρ_F leaves the local sign unchanged.

The shift is essential. With $\beta = 0$, the quadratic channel is locally even around the stationary centre and does not generate a stable directional leverage effect. The fast bank should not dominate total variance, because a large orthogonal or symmetric component dilutes correlation-based leverage statistics.

Status: P/N/O. The sign mechanism is explicit and has been observed in the programme’s simulations. Its empirical magnitude remains a calibration constraint.

4.5 Strong Zumbach effect

The Zumbach effect is a causal time-reversal asymmetry between past returns and future volatility. A representative statistic at horizon τ is

$$Z(\tau) = \text{Corr}(R_{t-\tau,t}^2, \text{RV}_{t,t+\tau}) - \text{Corr}(\text{RV}_{t-\tau,t}, R_{t,t+\tau}^2). \quad (24)$$

A reversible exogenous volatility process independent of returns has $Z(\tau) = 0$ in the ideal stationary limit. The current model breaks reversibility without making the OU driver amplitude depend on V :

1. the feedback state is a causal filter of a shock correlated with the return innovation;
2. the future variance depends quadratically on that state;
3. past return shocks therefore affect future variance through Y^2 , while the time-reversed pairing does not have the same causal construction.

The programme’s controlled Gate C experiment provides mechanism evidence:

Configuration	Pearson asymmetry
GARCH positive control	0.118 ± 0.009
Standardised correlated driver, quadratic readout	0.297 ± 0.011
Quadratic-ablation reference run	0.215 ± 0.018
Same ablation run with $q_Z = 0$	0.000 ± 0.002

These are replicated mechanism checks, not market calibration results. They show that the quadratic readout is load-bearing and that variance-dependent raw-return driving is not required. The standardised-driver sweep and the quadratic ablation used different reference

runs, so their absolute levels should not be compared as if they were one nested sample. This differs from endogenous price-feedback models such as quadratic rough Heston, but it targets the same time-reversal asymmetry [8, 6].

Measurement requirement

In the programme's validation, replacing realised variance by daily squared returns removed almost all power of the Zumbach statistic, even on a GARCH positive control. The empirical target should therefore use intraday realised variance, and every result should be accompanied by a reversible null, a positive control and replication uncertainty.

Status: N/O. The mechanism is strongly supported by ablation. Its market magnitude cannot be claimed before a correctly measured intraday-data study.

4.6 Heavy-tailed returns at relevant frequencies

Conditional on the variance path, short-horizon returns are approximately Gaussian; their unconditional distribution is a variance mixture. The exponential-quadratic channel produces large finite-frequency dispersion. In the scalar untempered illustration

$$\sigma = e^{qY^2}, \quad Y \sim \mathcal{N}(0, 1), \quad q > 0, \quad (25)$$

one has, up to a slowly varying logarithmic factor,

$$\mathbb{P}(\sigma > x) = \mathbb{P}\left(Y^2 > \frac{\log x}{q}\right) \asymp x^{-1/(2q)} (\log x)^{-1/2}. \quad (26)$$

This explains why an exponential quadratic can generate much heavier observable tails than a direct quadratic variance map. It does not, by itself, establish the tail index of a continuous-time return when the volatility state and return share Brownian increments.

For $X \sim \mathcal{N}(m, \Sigma)$ and a quadratic matrix Q , the untempered first variance moment requires

$$I - 4\Sigma^{1/2}Q\Sigma^{1/2} \succ 0. \quad (27)$$

Higher variance moments impose progressively stronger spectral conditions. The tempered link removes asymptotic moment explosion and replaces true power tails by a broad but ultimately truncated mixture. This is consistent with the stated objective: realistic heavy-tail behaviour over observable intraday and daily quantiles is more important than an exact asymptotic law.

The optional orthogonal block can increase log-volatility dispersion without changing the largest quadratic eigenvalue when its loading is spread over many independent directions and each new eigenvalue remains below the existing maximum. It should be activated only if the correlated feedback and fast banks cannot match return kurtosis and volatility quantiles without damaging leverage or Zumbach.

Status: P/N/O. Heavy finite-frequency tails are a strong capability of the readout. Their quantitative fit and stability across aggregation horizons remain empirical tests.

4.7 Taylor effect

The Taylor effect requires

$$\text{Corr}(|r_t|, |r_{t+k}|) > \text{Corr}(r_t^2, r_{t+k}^2) \quad (28)$$

over a relevant set of lags. It is not implied by leverage, roughness or clustering alone. For the benchmark representation $r_t = e^{X_t} \varepsilon_t$, where X is stationary Gaussian with variance s^2 , lag correlation ρ_k , and $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, direct calculation gives

$$\rho_{|r|}(k) = \frac{(2/\pi) (e^{s^2 \rho_k} - 1)}{e^{s^2} - 2/\pi}, \quad (29)$$

$$\rho_{r^2}(k) = \frac{e^{4s^2 \rho_k} - 1}{3e^{4s^2} - 1}. \quad (30)$$

At small s^2 , the first correlation grows more slowly, so the Taylor inequality fails. At $\rho_k = 0.9$, the crossover in this benchmark is approximately $s^2 = 0.081$, or $\text{sd}(\log \sigma) \approx 0.285$. The threshold changes with persistence and with departures from the Gaussian log-volatility benchmark.

The exponential-quadratic and optional multi-directional blocks provide the required dispersion more naturally than a direct quadratic variance readout. Nevertheless, Taylor must remain a diagnostic or low-weight target rather than a theorem of the architecture.

Status: P/N/O. The model can cross the analytically identified dispersion threshold; the sign and lag range must be checked after every joint calibration.

4.8 Fast volatility, downside asymmetry and VIX spike relaxation

Suppose a fast OU projection is shocked to $F_t = f_0$. Ignoring subsequent noise,

$$F_{t+s} \approx f_0 e^{-\lambda_J s}. \quad (31)$$

Its untempered contribution to relative variance is

$$\frac{V_{t+s}}{\bar{V}_{t+s}} \approx \exp\left(2q_J \left[f_0^2 e^{-2\lambda_J s} - 2\beta_J f_0 e^{-\lambda_J s}\right]\right), \quad (32)$$

where $\bar{V}$ uses the resting fast state $F = 0$. For equal-magnitude shocks $f > 0$,

$$\log \frac{V(F = -f)}{V(F = +f)} = 8q_J \beta_J f. \quad (33)$$

Thus $q_J > 0$, $\beta_J > 0$, and positive fast-bank/equity correlation make a negative equity shock produce a larger contemporaneous volatility response than an equal positive shock. The result is exact below the tempering region; the smooth cap can attenuate it in the far tail. The zero-shift restriction is the symmetric nested control.

The corresponding time derivative is

$$\frac{d}{ds} \log \left(\frac{V_{t+s}}{\bar{V}_{t+s}} \right) = -4\lambda_J q_J \left[f_0^2 e^{-2\lambda_J s} - \beta_J f_0 e^{-\lambda_J s} \right]. \quad (34)$$

For the downside shocks of interest ($f_0 < 0$), this derivative is negative and its magnitude is largest near the event before slowing progressively. Several fast and medium OU rates create a sharp front followed by a persistent shoulder. This temporal shape comes from the interaction of mean reversion with the exponential level map, not from a jump in the state process.

Fresh-seed mechanism screen

A paired screen of 16 stable Stage-3 M2 configuration–scenario pairs, five fresh seeds and six values $\beta_J \in \{0, 0.1, \dots, 0.5\}$ produced 480 evaluations. At $\beta_J = 0.1$, the downside-minus-upside median day-zero volatility-response gap increased from 35.0% to 38.7%, and the corresponding days 1–5 gap increased from 29.4% to 37.3%. All seven pre-existing Stage-3 proxy gates passed in 67 of 80 evaluations; six of 16 configurations passed on every fresh seed. All failures at this shift came from the positive-Zumbach lower bound. This is controlled synthetic mechanism evidence, not an empirical SPX calibration.

The cap may flatten the very top of an extreme spike. Its level should be selected so that it almost never binds in the calibration region while still controlling martingality outside it.

Status: P/N/O. The relaxation shape follows from the model and the downside orientation has controlled numerical support. Matching the distribution, amplitude and equity co-movement of observed VIX spikes is an empirical requirement.

4.9 Summary of physical-measure support

Target	Status	Load-bearing component
Near-zero return ACF	P/O	Diffusive return innovation and parsimonious deterministic Sharpe ratio
Volatility clustering	C/P/O	Slow exogenous OU mixture
Finite-scale roughness	P/O	Shared-driver rough OU lift and appropriate rate grid
Leverage	P/N/O	Shift β , curvature q_Z , and return correlation ρ_F
Strong Zumbach	N/O	Causal correlated OU filter plus quadratic readout
Heavy finite-frequency tails	P/N/O	Exponential quadratic; optional orthogonal dispersion block
Taylor effect	P/N/O	Sufficient log-volatility dispersion and persistence
Fast asymmetric VIX-like spikes	P/N/O	Shifted fast, partly correlated OU projection in the exponential quadratic

5 Risk-neutral pricing and joint SPX–VIX calibration

5.1 Forward variance, VIX and VIX futures

By equation (10), the input curve satisfies

$$\mathbb{E}_0^{\mathbb{Q}}[V_t] = \xi_0(t). \quad (35)$$

For a VIX window $\Delta = 30/365$, ignoring the conventional discrete-sampling and corridor adjustments,

$$\text{VIX}_T^2 = \frac{100^2}{\Delta} \int_T^{T+\Delta} \mathbb{E}_T^{\mathbb{Q}}[V_u] du. \quad (36)$$

Matching ξ_0 does not automatically match VIX futures, because

$$F_0^{\text{VIX}}(T) = \mathbb{E}_0^{\mathbb{Q}}[\text{VIX}_T] \leq \sqrt{\mathbb{E}_0^{\mathbb{Q}}[\text{VIX}_T^2]}. \quad (37)$$

The difference is the VIX convexity adjustment and must be calibrated or reported, not hidden inside the forward-variance normalisation.

5.2 Analytic Gaussian-core moments

If $\delta = 0$ and $h(x) = x$, the OU state is conditionally Gaussian and the score is quadratic. If

$$X_u \mid \mathcal{F}_T \sim \mathcal{N}(m_{T,u}, \Sigma_{T,u}) \quad (38)$$

and the exponent is $X^\top A X + b^\top X + c$, then

$$\mathbb{E}_T \left[e^{X^\top A X + b^\top X + c} \right] \quad (39)$$

$$= \frac{\exp\left(c + m^\top A m + b^\top m + \frac{1}{2}(2A m + b)^\top (I - 2\Sigma A)^{-1} \Sigma (2A m + b)\right)}{\sqrt{\det(I - 2\Sigma A)}}, \quad (40)$$

provided

$$I - 2\Sigma^{1/2} A \Sigma^{1/2} \succ 0. \quad (41)$$

This turns equation (36) into a one-dimensional deterministic maturity quadrature evaluated at the cached state X_T . No nested Monte Carlo is needed in the Gaussian shadow model.

5.3 Tempering and the nonlinear ESN layer

The recommended tempered model does not retain the closed form equation (40). Neither does the full model with $\delta \neq 0$. It remains computationally tractable through one of the following offline–online decompositions:

1. low-dimensional Gauss–Hermite or sparse-grid quadrature when the active quadratic directions are few;
2. functional quantisation of the Gaussian state;
3. least-squares conditional expectation on cached states;
4. a neural surrogate trained on the state and parameter domain, with the Gaussian core used as a control variate.

Quantisation and neural pricing have already been demonstrated for Gaussian polynomial SPX–VIX models [2]; precomputed signature samples provide another successful offline calibration paradigm [4]. These are useful numerical templates, not proofs that the present model will calibrate.

5.4 SPX pricing and martingality

For fixed cached innovations and a candidate parameter vector, compute

$$\log \frac{S_T}{S_0} = \int_0^T \left(r_t - d_t - \frac{1}{2} V_t \right) dt + \int_0^T \sqrt{V_t} dW_t^{S, \mathbb{Q}}. \quad (42)$$

The discounted stock is always a nonnegative local martingale. It is a true martingale if the stochastic exponential has expectation one. In the tempered model, V is bounded on every finite horizon when ξ_0 and the deterministic risk premia are bounded and the positive normaliser $M_{\mathbb{Q}}$ is bounded away from zero. Novikov's condition then gives

$$\mathbb{E}^{\mathbb{Q}} \left[\exp \left(\frac{1}{2} \int_0^T V_t dt \right) \right] < \infty, \quad (43)$$

so the discounted stock is a true martingale. In the untempered, return-correlated exponential-quadratic model, finite marginal moments do not by themselves establish true martingality; the admissible domain must be analysed separately.

Production choice

The smooth upper tempering is part of the recommended model, not merely a numerical emergency stop. It provides a clean finite-horizon martingale argument while preserving the untempered map over the empirically relevant state region.

5.5 Why a quintic polynomial is not structurally required

The fifth-degree polynomial used by Abi Jaber, Illand and Li is effective in their one-factor Gaussian polynomial family and helps generate upward VIX smiles [2]. It is not a necessity theorem. Two complementary results make this clear:

- quadratic rough Heston combines a quadratic variance map with rough feedback and provides a continuous-path solution to the joint SPX–VIX smile problem [6];
- signature models obtain strong joint calibrations using several low-order path features rather than a scalar quintic readout [4].

The present architecture substitutes feature diversity, rate diversity and exponential convexity for scalar polynomial degree. Its fast and slow quadratic directions have distinct economic roles, which should improve identifiability relative to an unrestricted high-degree polynomial.

6 Joint calibration programme

6.1 Recommended nested sequence

Model	Readout	Purpose
M0	Positive direct quadratic variance	Minimal baseline; tests whether exponential convexity is genuinely needed.
M1	Untempered Gaussian exponential quadratic, $\delta = 0$	Analytic shadow model; moment-domain and VIX-control-variate benchmark.
M2	Tempered Gaussian exponential quadratic, $\delta = 0$	Recommended production core.
M3	M2 plus $\delta^\top Z$	Activate only if the nonlinear ESN layer materially improves out-of-sample physical facts or option errors.
M4	M3 plus orthogonal block $q_O > 0$	Optional finite-frequency dispersion and tail tuning; activate only after diagnosing a residual deficiency.

If M2 cannot reproduce synthetic surfaces generated by a quintic OU benchmark, add a second identified quadratic direction before increasing polynomial degree. This directly tests the hypothesis that factor diversity can replace degree.

6.2 Relation of M2 to ESN–A2

M2 is a structured descendant of ESN–A2, but it is neither a renaming nor a nested restriction of A2. The common core is a fixed multirate Gaussian reservoir whose state can be reused while the volatility readout changes. The redesign concerns the organisation of that state, the readout parameterisation, and the separation between physical and pricing inputs. The comparison below refers to the supplied reconstructed A2 engine; its implementation file records that it was reconstructed after an environment reset and validated against the reported roughness targets. Consequently, code-specific A2 conclusions carry that provenance.

On a grid of width Δ , the A2 rough bank and nonlinear layer can be written schematically as

$$r_{n+1} = D_\lambda r_n + c_\lambda \varepsilon_{n+1}, \quad D_{\lambda,jj} = e^{-\lambda_j \Delta}, \quad c_{\lambda,j} = \sqrt{1 - e^{-2\lambda_j \Delta}}, \quad (44)$$

$$z_{n+1} = \Phi_{A2}(r_{n+1}, z_n), \quad (45)$$

where the same scalar innovation ε_{n+1} drives every coordinate of r and the equity return in the supplied simulator. The map Φ_{A2} is the leaky, bounded second layer built from quadratic and linear features, a hyperbolic tangent, local recurrence, and the state-dependent norm gate. Its volatility score and variance map are

$$\eta_n^{A2} = b_0 + \kappa q^\top r_n + w_z^\top z_n + \frac{m_1}{N_r} \|r_n\|^2 + \frac{m_2}{N_r} (q^\top r_n)^2, \quad (46)$$

$$V_n^{A2} = \sigma_{\min}^2 + \text{softplus}(\eta_n^{A2})^2. \quad (47)$$

At the A2 resting bias, softplus is already close to its exponential branch. Thus A2 already contains the central idea ‘‘Gaussian reservoir plus approximately exponential-quadratic readout.’’ In contrast, M2 sets $\delta = q_O = 0$ in equation (8) and uses

$$\eta_t^{M2} = c_C^\top X_t^C + q_Z(Y_t - \beta)^2 + q_J(F_t - \beta_J)^2, \quad (48)$$

$$V_t^{M2} = \underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t^{M2})}{M_Q(t)}. \quad (49)$$

The models therefore share a mathematical backbone, but they are not the same parameterised family:

Component	ESN–A2	M2
State drivers	One multirate rough bank driven by the equity innovation; the coupling is effectively perfect in the supplied engine.	Separate feedback, fast-spike, and clustering banks. Feedback and spike shocks may be only partially correlated with the equity shock; clustering shocks are independent.
Nonlinear layer	The bounded recurrent z -layer contributes directly through $w_z^\top z$.	Inactive: $\delta = 0$. M3, not M2, restores an optional fixed contractive echo layer.
Leverage and Zumbach	The linear rough projection, aligned quadratic projection, and nonlinear layer have overlapping effects, so attribution is parameter-regime dependent.	Concentrated in the shifted correlated channel $q_Z(Y - \beta)^2$, whose slope controls the leverage sign and whose state dependence generates time-reversal asymmetry.
Clustering	Encoded jointly by the multirate bank, especially its isotropic reservoir-energy term, and persistence transmitted through the nonlinear layer.	A dedicated slow linear bank $c_C^\top X^C$, separated from the leverage–Zumbach channel.

Component	ESN–A2	M2
Fast VIX-like spikes	No separately identified fast-spike bank.	A dedicated convex direction $q_J(F - \beta_J)^2$, with its own fast rate grid and return correlation; $\beta_J > 0$ orients equal-magnitude responses toward downside shocks.
Variance map and tails	Squared softplus with an unbounded exponential branch; the quadratic eigenvalues govern its asymptotic moment domain.	A direct exponential-quadratic shadow map, smoothly tempered in production and close to the shadow map below the cap. The target is finite-frequency fat tails, not an asymptotic power law.
Measure change	The supplied engine is a physical-measure simulator and does not give a complete, separately normalised $\mathbb{P}/\mathbb{Q}$ specification.	Explicit $\mathbb{P}$ and $\mathbb{Q}$ stock equations, deterministic Girsanov kernel, unspanned risk premia, and a market forward-variance normalisation.
Path reuse	Available while parameters that determine r and z are fixed; changing reservoir or second-layer parameters rebuilds the state.	Primitive shocks and exogenous states are cached. Readout, forward-curve, and deterministic risk-premium changes require deterministic state adjustments and revaluation, but no new random draw.
Calibration roles	The audited parameter split allowed quantities intended for daily recalibration to move most of the physical statistics.	Architecture, shared structural parameters, measure-dependent premia, and daily inputs are separated explicitly; shared readout coefficients cannot be overwritten by a daily $\mathbb{Q}$ fit.

The shifted M2 channel makes the relationship particularly transparent:

$$q_Z(Y - \beta)^2 = q_Z Y^2 - 2q_Z \beta Y + q_Z \beta^2. \quad (50)$$

It combines A2’s aligned linear and quadratic rough-factor channels, but imposes the identified coefficient relation $\kappa_Y = -2q_Z \beta$ after normalisation. Conversely, baseline M2 drops A2’s isotropic energy and nonlinear readouts, replaces their clustering role by X^C , and adds the fast direction F . M4 can restore an independent quadratic dispersion channel, but that channel lives in an orthogonal bank and is therefore not equivalent to A2’s $\|r\|^2$. M3 restores nonlinear memory, but its controlled echo layer does not make the resulting model identical to A2 either.

Nesting statement

A2 is not a parameter point of M2, and M2 is not a parameter point of A2. M2 is best viewed as a lower-nonlinearity but more economically structured redesign of A2. It retains the exogenous Gaussian and exponential-quadratic mechanisms, while changing the driver topology, assigning separate channels to clustering and spikes, enforcing path-reuse discipline, and adding the objects required for joint SPX–VIX pricing.

The transfer of stylised-fact evidence is therefore conditional. A2 experiments motivate the

rough-reservoir and convex-readout mechanisms, but they do not prove that M2 preserves their empirical magnitudes. M2 must be evaluated on its own with matched estimators, multiple seeds, and the ablations prescribed below. Likewise, M2 has better specified joint-pricing capacity, not a demonstrated SPX–VIX fit in the absence of option data.

6.3 Calibration hierarchy

Let Θ_H , Θ_S , Θ_M and D_t denote architectural hyperparameters, shared structural parameters, measure-dependent parameters and daily inputs. A practical hierarchy is:

1. **Architecture selection.** Choose dimensions and rate grids using rolling physical-measure validation and synthetic option-capacity tests. This is the expensive outer loop.
2. **Shared structural estimation.** Estimate Θ_S from historical windows and a panel of option surfaces, with stability penalties across dates.
3. **Measure change.** Estimate the thin deterministic unspanned risk premia in Θ_M , with θ pinned by the physical drift convention.
4. **Daily calibration.** Strip ξ_0 , filter the current state and calibrate only the permitted daily pricing quantities.

Because shared readout parameters affect both $\mathbb{P}$ and $\mathbb{Q}$, a purely sequential "fit $\mathbb{P}$, then overwrite under $\mathbb{Q}$ " procedure is not valid. Use either a joint loss or alternating minimisation with a physical-measure retention penalty:

$$\mathcal{L}(\Theta) = w_P \mathcal{L}_P + w_S \mathcal{L}_{\text{SPX}} + w_V \mathcal{L}_{\text{VIX}} + w_F \mathcal{L}_{\text{VIX futures}} + \mathcal{R}(\Theta). \quad (51)$$

Option residuals should be expressed in bid–ask-normalised prices, for example

$$\varepsilon_i = \frac{C_i^{\text{model}} - C_i^{\text{mid}}}{\max\{\frac{1}{2}(C_i^{\text{ask}} - C_i^{\text{bid}}), \epsilon_i^{\text{floor}}\}}, \quad (52)$$

rather than unweighted implied-volatility errors. VIX futures must be included separately from the forward-variance curve.

6.4 Validation and falsification tests

Question	Required test	Failure implication
Roughness	Structure functions on noise-robust intraday realised volatility across the designed lag band	Retune rate grid; do not add readout degree.
Leverage	Lagged return–realised-volatility correlation with confidence bands	Adjust β , ρ_F , or add a separate linear leverage bank.
Zumbach	Realised-variance statistic, reversible null, positive control, multiple seeds or blocks	Reassess the correlated quadratic mechanism or measurement protocol.

Question	Required test	Failure implication
Heavy tails	QQ/body fit, kurtosis, threshold-stability plots and aggregation horizons	Add orthogonal dispersion only after excluding measurement and cap effects.
Taylor	Both correlation curves and the log-volatility dispersion threshold	Treat as secondary unless failure contaminates more important targets.
VIX spikes	Peak amplitude, rise/decay asymmetry and SPX co-movement	Retune fast rates and ρ_J ; avoid a dominant independent spike factor.
Martingale	Analytic boundedness proof plus Monte Carlo defect within sampling error	Reject parameter point or lower the cap.
Joint surface capacity	Bid-ask-normalised SPX/VIX errors and VIX-futures errors over dates	Add a second quadratic direction before quintic degree.
Path reuse	Bitwise equality of cached X, Z under readout perturbations	Implementation violates the architectural constraint.

7 Open issues and limitations

What is not yet established

1. No SPX/VIX option data have yet been used, so joint calibration capacity is untested.
2. Identifiability of curvature, shift, correlations, fast rates and deterministic risk premia must be studied through Jacobian singular values, profile likelihoods and rolling stability.
3. The optimal cap level must balance negligible in-sample binding against robust moment and martingale control.
4. A finite OU lift produces only finite-scale roughness.
5. The exact market magnitude of the Zumbach effect requires intraday realised variance; daily squared returns are an inadequate proxy for the current statistic.
6. The nonlinear second layer is not currently justified by the Gate C Zumbach ablation. It must earn its parameters through out-of-sample improvement.

Three interactions deserve particular monitoring:

- Increasing an independent dispersion block can improve tails and Taylor while diluting leverage and Zumbach correlations.
- Increasing the correlated quadratic curvature strengthens Zumbach and tails but pushes the untempered model toward moment and martingale boundaries.
- A low cap improves safety but can flatten extreme VIX spikes, distort roughness estimates in the upper tail and limit the VIX smile.

These are not arguments against the model. They are the principal trade-offs that the joint objective must expose rather than hide.

8 Conclusion

The current candidate is a two-layer exogenous reservoir architecture with a recommended Gaussian first calibration and an optional nonlinear ESN extension. Its defining choices are:

- multirate OU banks with variance-independent drivers;
- a return-correlated rough feedback projection;
- a shifted quadratic feedback term for leverage and Zumbach;
- slow exogenous modes for clustering and path reuse;
- a shifted fast, partly return-correlated direction for downside-oriented VIX spike relaxation, nested at $\beta_J = 0$;
- a normalised, smoothly tempered exponential-quadratic variance map;
- a deterministic market price of risk preserving the Gaussian OU class under $\mathbb{Q}$;
- an optional independent dispersion block and optional nonlinear ESN layer, both subject to ablation.

This construction retains the mechanisms needed for the targeted physical-measure facts and offers a viable offline–online route to SPX/VIX calibration without a quintic polynomial. Roughness and path reuse are architectural; leverage has an explicit sign condition; Zumbach has controlled numerical support; tails and VIX spikes follow from exponential-quadratic convexity; and Taylor follows only above a measurable dispersion threshold.

The next decisive step is not another architectural elaboration. It is a disciplined implementation of M0–M2, validation on intraday physical data, and synthetic joint-calibration tests against quadratic-rough-Heston and quintic-OU benchmarks before real option data become available.

A Derivation of the Gaussian quadratic transform

Let $X = m + Z$, with $Z \sim \mathcal{N}(0, \Sigma)$, and let $A = A^\top$. Write

$$X^\top AX + b^\top X + c = c + m^\top Am + b^\top m + u^\top Z + Z^\top AZ, \quad u = 2Am + b. \quad (53)$$

Whitening with $Z = \Sigma^{1/2}U$ and completing the square in the Gaussian density gives

$$\mathbb{E}\left[e^{u^\top Z + Z^\top AZ}\right] = \det(I - 2\Sigma A)^{-1/2} \exp\left[\frac{1}{2}u^\top (I - 2\Sigma A)^{-1}\Sigma u\right], \quad (54)$$

provided $I - 2\Sigma^{1/2}A\Sigma^{1/2} \succ 0$. Substitution yields equation (40). For $e^{2\eta}$ with quadratic matrix Q , set $A = 2Q$, which gives the condition in equation (27).

B Derivation of the Taylor benchmark

Let $r_t = e^{X_t} \varepsilon_t$, where $\mathbb{E}[X_t] = \mu$, $\text{Var}(X_t) = s^2$, $\text{Corr}(X_t, X_{t+k}) = \rho_k$, and the standard-normal innovations are independent through time and independent of X . Then

$$\mathbb{E}[e^{X_t}] = e^{\mu+s^2/2}, \quad \mathbb{E}[e^{2X_t}] = e^{2\mu+2s^2}, \quad (55)$$

$$\mathbb{E}[e^{X_t+X_{t+k}}] = e^{2\mu+s^2(1+\rho_k)}, \quad \mathbb{E}[\varepsilon] = \sqrt{2/\pi}. \quad (56)$$

Using these moments in the covariance and variance of $|r|$ yields the first formula in equation (30). Applying the same calculation to r^2 , with $\mathbb{E}[\varepsilon^4] = 3$, gives the second. The result is a diagnostic approximation for the current nonlinear model, not an exact formula for it.

C Related model classes

Class	Main device	Relation to the present model
Gaussian polynomial volatility	Polynomial of a Gaussian Volterra state	Demonstrates strong joint SPX–VIX calibration and motivates Gaussian path reuse; degree five is specific to that one-factor family.
Quadratic rough Heston	Endogenous rough price feedback and quadratic variance	Shows that a quadratic map can support strong Zumbach and joint smiles, but sacrifices exogenous frozen states.
Signature volatility	Linear readout of rich path features	Demonstrates that feature diversity and offline samples can replace high scalar polynomial degree.
Rough-volatility lift	Exponential-sum approximation of fractional kernels	Supplies the multirate Markovian mechanism used for finite-scale roughness.
Classical ESN	Fixed contractive recurrent reservoir plus trained readout	Supplies the optional nonlinear second layer and the path-reuse philosophy.

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